Grade 6 Accelerated
Unit 2
Lesson 7 Homework

Name_Answer Key__________
Date______________________
Period_____________________

1. Raechelle is making lemonade to bring to a cookout. She mixes $\frac{3}{4}$ of a cup of sugar, $1\frac{1}{8}$ cups of lemon juice, and $6\frac{1}{2}$ cups of water. If Raechelle serves each person $\frac{1}{2}$ a cup of lemonade, how many servings will she have?

   $\left(\frac{3}{4}+1\frac{1}{8}+6\frac{1}{2}\right)\div\frac{1}{2}\rightarrow$ $\left(\frac{6}{8}+1\frac{1}{8}+6\frac{4}{8}\right)\div\frac{1}{2}\rightarrow$ $8\frac{3}{8}\div\frac{1}{2}\rightarrow\frac{67}{8}\div\frac{1}{2}\rightarrow$ $\frac{67}{8}\times\frac{2}{1}=\frac{67}{4}$ or $16\frac{3}{4}$ servings

2. At Gamestop, Andreas bought 3 games that cost $\$19.75$ each. He had a coupon for $\$15$ off and paid an additional $\$4.87$ in tax. What was the total amount that Andreas spent at Gamestop?
   $19.75\times 3=\$59.25$ $59.25-15.00=\$44.25$ $44.25+4.87=\$49.12$

   He paid $\$49.12$ total

3. Simeon is learning to speak Spanish. He practices Spanish for $\frac{3}{5}$ of an hour for 4 days and then practices for $1\frac{1}{4}$ hours for the next 3 days. How many total hours did he practice over the 7 days?

   $$\left(4\times\frac{3}{5}\right)+\left(3\times 1\frac{1}{4}\right)\rightarrow\left(\frac{4}{1}\times\frac{3}{5}\right)+\left(\frac{3}{1}\times\frac{5}{4}\right)=\frac{12}{5}+\frac{15}{4}\rightarrow\frac{48}{20}+\frac{75}{20}=\frac{123}{20}$$

   $\frac{123}{20}=6\frac{3}{20}$ or 6 hours and 9 minutes practicing.

4. Hazel is going to the movies with her friends. She mixed 4.9 ounces of kettle corn, 6.42 ounces of caramel corn, and 5.83 ounces of cheddar popcorn. If she splits the popcorn mix evenly into three containers, how many ounces of the mix will be in each container?

   $$\begin{array}{r} 6.42 \\ 5.83 \\ +\ 4.90 \\ \hline 17.15 \end{array}$$

   $$\begin{array}{r} 5.71\overline{6} \\ 3\,\overline{)\,17.150} \\ -15 \\ \hline 21 \\ -21 \\ \hline 05 \\ -\ 3 \\ \hline 20 \end{array}$$

   $5.71\overline{6}$ ounces in each container

5. Suppose Hazel also includes 2.61 ounces of butter popcorn in her mix before dividing it into three containers. Now how many ounces will be in each container?

   $$\begin{array}{r} 17.15 \\ +\ 2.61 \\ \hline 19.76 \end{array}$$

   $$\begin{array}{r} 6.58\overline{6} \\ 3\,\overline{)\,19.76} \\ -18 \\ \hline 17 \\ -15 \\ \hline 26 \\ -\ 24 \\ \hline 20 \end{array}$$

   $6.58\overline{6}$ ounces in each container

6. Solange delivers packages on her bike. She biked $8\frac{2}{3}$ miles North, $7\frac{3}{5}$ miles East, and another $6\frac{1}{2}$ miles North. If she biked the same way back to her starting point, how many total miles did she bike round-trip?

$2\left(8\frac{2}{3}+7\frac{3}{5}+6\frac{1}{2}\right)=2\left(8\frac{20}{30}+7\frac{18}{30}+6\frac{15}{30}\right)=2\left(21\frac{53}{30}\right)=2\left(22\frac{23}{30}\right)=45\frac{8}{15}$ total miles

7. Mary had $\$16.15$ from delivering newspapers and $\$11.50$ from her allowance. She takes all her money to a carnival. Admissions to the carnival is $\$5.25$ and rides cost $\$1.40$ each. How many rides can Mary go on after paying the admissions fee?

$16.15+11.50=\$27.65-5.25=\$22.40$

$\frac{22.40}{1.40}=16$ → she can go on 16 rides

8. Elliot is making a layer cake. He needs $1\frac{1}{4}$ cups of sugar for the cake, $\frac{1}{2}$ cup of sugar for the filling, and $\frac{3}{4}$ cup of sugar for the icing. If Elliot only has a $\frac{1}{8}$ cup scoop, how many total scoops of sugar will he use?

$\left(1\frac{1}{4}+\frac{1}{2}+\frac{3}{4}\right)\div\frac{1}{8}=\left(1\frac{1}{4}+\frac{2}{4}+\frac{3}{4}\right)\div\frac{1}{8}=1\frac{6}{4}\div\frac{1}{8}=2\frac{1}{2}\div\frac{1}{8}=\frac{5}{2}\times\frac{8}{1}=20$ scoops of sugar

Maili runs at a pace of 8.2 minutes per mile. Her brother's pace is 9.5 minutes per mile.

9. If Maili and her brother complete a relay race in which they each run 2.5 miles, what will be their total time to complete the race?

Maili → 8.2×2.5 miles $=20.5$ minutes
Brother → 9.5×2.5 miles $=23.75$ minutes
$20.5+23.75=44.25$ minutes total

10. Maili and her brother are both planning to do a 5 kilometer race. 1 kilometer ≈ 0.62 miles. How many minutes before her brother will Maili finish?

Maili → 8.2 min/mile

Brother → 9.5 min/mile

$\therefore 1$ km$=$ 0.62 miles

5 km$=$ x

$\therefore x = \frac{5\ km \cdot 0.62\ miles}{1\ km} = 3.1$ miles

$\therefore 8.2$ min/mile $\times\, 3.1$ miles $= 25.42$ minutes

9.5 min/mile $\times\, 3.1$ miles $= 29.45$ minutes

$\therefore 29.45 - 25.42 = 4.03$ minutes

Maili will finish 4.03 minutes before her brother.